Madan Bhandari University of Science and Technology
Open Assessment 1

Program: Digital Technology
Subjects: AI-CR-601 (Advanced Deep Learning)
Level: Masters In Applied Science

Semester: III
Submission deadline:
20 September 2026, 5 pm

Answer all questions **Total Marks: 50**

1. [8 Marks] Implement Recurrent Neural Network and LSTM from scratch using pytorch framework. Using your LSTM implementation to create a Next word prediction (4-16 sequence length) model. Provide evaluation results on unseen test data.
2. [8 Marks] Replace the default Recurrent block with your own vanilla Transformer implementation.
3. [10 Marks] Generate different prompts (fill in the blanks, translation, next word) and using a dataset, and these prompts, create a prompt-able transformer model on these different kinds of tasks.
4. [24 Marks] Use OPUS/NLLB nepali-english translation dataset, use same tokenizer and embedding for english and nepali (maybe gemma, qwen). Use a cactus transforme model and train an encoder-decoder translation model. Experiment with passing encoder embeddings to 1 layer vs all layers. You only need to train on smallerr sample. Provide evaluation results on FLORES eng-nep and nep-eng ***training dataset***.